Copernicus Marine Service

QUALITY INFORMATION DOCUMENT

Ocean Colour Production Centre

Baltic Sea Observation Products
OCEANCOLOUR_BAL_BGC_L3_NRT_009_131
OCEANCOLOUR_BAL_BGC_L4_NRT_009_132
OCEANCOLOUR_BAL_BGC_L3_MY_009_133
OCEANCOLOUR_BAL_BGC_L4_MY_009_134

Issue: 2.0

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CHANGE RECORD

When the quality of the products changes, the QuID is updated and a row is added to this table. The third column specifies which sections or sub-sections have been updated. The fourth column should mention the version of the product to which the change applies.

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2.0	15/07/ 2024	all	Updates for EIS November 2024	V.E. Brando, L. Gonzalez Vilas, A. Di Cicco, M. Sammartino, S. Colella, D D'Alimonte, T. Kajiyama, S. Kaitala, J. Attila	V.E. Brando

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I EXECUTIVE SUMMARY

I.1 Products covered by this document

This document covers the Copernicus Marine Service Baltic (BAL) regional observation datasets for the near-real time (NRT) and reprocessed (MY) products. Table 1 show the list of products available through the Ocean Colour Thematic Assembly Centre (OCTAC) of the Copernicus Marine Service (CMEMS), the scientific validation of which is provided in this document.

The BAL chlorophyll-a and the water reflectance datasets based on Sentinel-3 OLCI at ~300m spatial resolution are produced both in NRT and in MY (from 2016 onwards) based on the EUMETSAT OLCI V3.0.1 reprocessing completed in 2022 as the daily Level-3 (L3) and monthly Level 4 (L4) (Table 1, Table 2).

The BAL 1997-2023 reprocessed MY time series includes surface chlorophyll concentration (CHL), both as daily L3 and monthly L4, and daily L3 remote-sensing reflectances (Rrs) at ~1 km spatial resolution (Table 1, Table 2). This subset of remote sensing reflectances (bands 412-443-490-510-555-670) were directly extracted from ESA OC-CCI (European Space Agency Ocean Colour Climate Change Initiative) Rrs input daily composites, provided for Copernicus Marine Service by the Plymouth Marine Laboratory (PML), which uses the OC-CCI processor (version 6.0[1]) to merge at high resolution MERIS, MODIS-AQUA, SeaWiFS, VIIRS and OLCI data. These data are remapped over the Baltic grid and repacked for compliance with the CMEMS format standards. The global quality information of the input data is presented in OC-CCI global MY QUID (CMEMS-OC-QUID-009-107to108[2]).

Table 1: List of products for which the scientific validation info is provided in this document

EPST name (OC-CNR-ROME-IT)	Zone/resolution	Code/Variable	Availability	Validation
OCEANCOLOUR_BAL_BGC_L3_NRT_009_131	BAL/ 0.3 km	plankton/CHL plankton/PFT reflectance/RRS transp/KD	Daily	Offline
OCEANCOLOUR_BAL_BGC_L4_NRT_009_132	BAL/ 0.3 km	plankton/CHL	Monthly	Offline
OCEANCOLOUR_BAL_BGC_L3_MY_009_133	BAL/ 0.3 km and 1 km	plankton/CHL plankton/PFT reflectance/RRS transp/KD	Daily	Offline
OCEANCOLOUR_BAL_BGC_L3_MY_009_134	BAL/ 0.3 km and 1 km	plankton/CHL	Monthly	Offline

Table 2: Overview of the near real-time (NRT) and reprocessed (MY) products for the Baltic Sea: OCEANCOLOUR_BAL_BGC_L3/L4_NRT/MY_009_131-134.

	multi sensor - 1km				Sentinel-3 OLCI A+B - 300m			
	NRT		MY		NRT		MY	
	L3	L4	L3	L4	L3	L4	L3	L4
BAL	-	-	133	134	131	132	133	134

[1] www.esa-oceancolour-cci.org
[2] https://catalogue.marine.copernicus.eu/documents/QUID/CMEMS-OC-QUID-009-107to108.pdf

For both NRT and in MY products, CHL is retrieved by an ensemble Muli Layer Percepetron (MLP) algorithm developed for the Baltic Sea. In this version, the ensemble scheme was reformulated using the Chi-squared distribution function to allow the introduction of a validity flag based on a confidence level. This product also includes the Phytoplankton Functional Types (PFT) dataset. PFT is calculated by applying regional algorithms developed for the Baltic Sea to the BAL MY CHL dataset and includes the following variables, expressed as Chlorophyll a concentration in sea water: Micro (Micro-phytoplankton), Nano (Nano-phytoplanton) and Pico (Pico-phytoplankton), also known as "Phytoplankton Size Classes" (PSCs), Diato (Diatoms), Dino (Dinophytes or Dinoflagellates), Crypto (Cryptophytes), Green (Green algae and Prochlorophytes) and Prokar (Prokaryotes). More details on the phytoplankton groups are provided in section II.7.2. From this version, the cyanobacterial subsurface and surface summer blooms are detected by applying the thresholds on R_{rs} at the wavelength of 555 and 670 nm as detailed in section II.7.3.

The scientific validation framework for the BAL area is also described in this document. The offline validation corresponds to the assessment of basic statistical quantities from the comparison of satellite-derived products and their *in situ* counter parts, when available. The validation of the OC NRT and MY Rrs datasets was conducted with *in situ* data collected at AERONET-OC (Zibordi et a;., 2009) stations. The validation of the OC NRT and MY Chlorophyll datasets was conducted with *in situ* data collected from several institutions in the framework of HELCOM and which are compiled in the COMBINE database, as described in section III.

I.2 Summary of the results

Estimated Accuracy Numbers (EANs) (Table 3) provide a quick overview of the product performance with respect to *in situ* observations. All EANs are defined in section III.1 and the equations for the metrics are provided in Table 7. The overall products presented in this document present an acceptable quality in terms of both their temporal consistency and with respect to their *in situ* counterparts in these optically complex waters.

The performance of the updated Ensemble MLP Chlorophyll algorithm show a Determination coefficient of 0.312 and Mean Relative Percentage Difference (RPD) and Mean Absolute Percentage Difference (APD) of -2.8 and 66.1% for the MY multi sensor datasets, and a Determination coefficient of 0.324 and RPD and APD of 4 and 51.2% for the NRT ad MY OLCI 300m datasets. The EANs for the OLCI results are consistent with the multi sensor datasets even if there was a lower number of matchups available from 2016 to date (460 vs 2070).

The performance of the Phytoplankton Functional Types and Phytoplankton Size Classes introduced in 2021 was assessed only for the multi sensor datasets due to the limited *in situ* data. The EANs for the PSCs and PFTs are consistent with those of the CHL retrievals.

The validation of the radiometric products was not updated in this version of the document. The radiometric products are validated against *in situ* Rrs at three AERONET-OC sites. The CCIv6.0 Rrs used in the multi-sensor MY datasets show an adequate agreement with *in situ* data in such optically complex waters in terms of correlation for the 443-665 nm spectral range (Determination coefficient 0.80-0.90), while the 412 and 442 nm bands show higher uncertainties (Determination coefficient of 0.42 and 0.62). The OLCI Rss used in the NRT ad MY OLCI 300m datasets was adequate in terms of correlation for the 443-665 nm spectral range (Determination coefficient 0.62-0.89) and relative and absolute percentage differences for the 490-665 nm spectral range (APD and RPD: 11-30%). For both datasets, a positive bias is notable: for OC-CCI in the 412-510 nm spectral range (bias ranging 2-6 10^{-5} sr^{-1}), with only the 560 and 665 nm achieving a low bias of ~6 10^{-5} sr^{-1}, and in OLCI across the whole

spectral range (bias ranging 1-6 10^{-4} sr^{-1}).The 400, 412 and 779 nm spectral bands are not used as input for the retrieval of any of the derived products, so the higher uncertainties can be deemed acceptable.

I.3 Estimated Accuracy Numbers

Table 3: Estimated Accuracy Numbers as defined in section III.1. Full metrics provided in Table 7. r^2: coefficient of determination; Root Mean Square Difference (RMSD); Centre-pattern Root Mean Square Difference (cRMSD); Bias; Mean Relative Percentage Difference (RPD); Mean Absolute Percentage Difference (APD).

Sensor	Variable	Mode	r^2	cRMSD	Bias	RPD	APD	N
MULTI	CHL	MY	0.312	0.375	0.335	-2.8	66.1	2070
OLCI	CHL	MY	0.324	0.271	0.262	4.0	51.2	460
MULTI	Micro	MY	0.413	0.319	0.065	55.3	84.7	192
MULTI	Nano	MY	0.179	0.489	0.027	109.3	153.9	191
MULTI	Pico	MY	0.238	0.311	-0.160	-6.5	58.3	192
MULTI	Diato	MY	0.389	0.308	0.107	72.2	96.8	190
MULTI	Dino	MY	0.162	0.503	-0.111	34.2	93.4	145
MULTI	Crypto	MY	0.178	0.490	-0.058	72.1	127.8	191
MULTI	Green	MY	0.145	0.415	-0.023	43.8	93.2	191
MULTI	Prokar	MY	0.281	0.291	-0.068	7.7	54.5	182
OLCI	RRS400	MY	0.365	5.12E-4	-4.03E-6	22.7	143.9	557
OLCI	RRS412	MY	0.516	3.89E-4	3.22E-4	77.1	87.2	646
OLCI	RRS443	MY	0.658	3.05E-4	6.40E-4	75.2	75.5	646
OLCI	RRS490	MY	0.747	3.47E-4	4.37E-4	27.9	29.1	646
OLCI	RRS510	MY	0.809	3.55E-4	3.50E-4	17.3	18.9	557
OLCI	RRS560	MY	0.896	4.26E-4	3.22E-4	11.9	14.0	646
OLCI	RRS620	MY	0.783	2.93E-4	2.85E-4	26.8	28.7	557
OLCI	RRS665	MY	0.714	2.46E-4	1.10E-4	23.1	29.8	646
OLCI	RRS779	MY	0.254	3.44E-4	9.76E-4	1087.3	1727.6	557
MULTI	RRS412	MY	0.416	3.91E-4	5.34E-4	217.8	220.0	1095
MULTI	RRS443	MY	0.642	2.95E-4	6.10E-4	81.6	81.7	1095
MULTI	RRS490	MY	0.797	2.80E-4	2.96E-4	21.1	22.9	1095
MULTI	RRS510	MY	0.827	3.29E-4	2.50E-4	12.8	15.2	387
MULTI	RRS560	MY	0.902	3.46E-4	6.28E-5	3.7	9.7	1095
MULTI	RRS665	MY	0.781	1.64E-4	-5.86E-5	-5.5	16.3	1095

II PRODUCTION SYSTEM DESCRIPTION

II.1 Production Centre: Ocean Colour (OC) TAC

The Ocean Colour Thematic Assembly Centre (OCTAC) constitutes and operates the European Ocean Colour Service for the Copernicus Marine Service (CMEMS) providing global, pan-European and regional (Atlantic, Arctic, Baltic, Mediterranean, and Black Seas) ocean colour products (Figure 1). This document describes the main achievements of the validation activity performed over the ocean colour operational products. Details on the validation context are provided in the next section.
The OCTAC is a distributed centre made of four Production Units (PU), each of which in charge of the full production of single product. All products are then made available through the single Dissemination Unit (DU). Although managed centrally, each PU is responsible of the quality of its products.

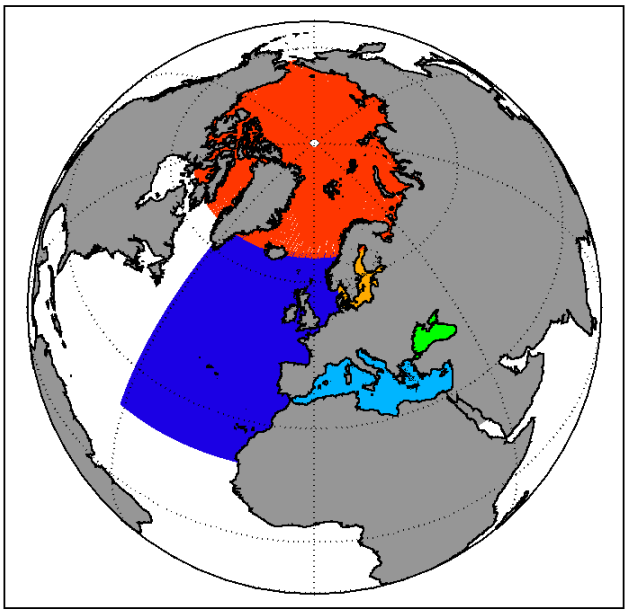


Figure 1: Graphical view of the CMEMS areas. The BAL region is in orange.

II.2 Production subsystem: OC-CNR-ROME-IT

In the OCTAC operations, the Italian National Research Council (CNR) provides near real-time and reprocessed mode products (Table 1, Table 2). This document provides detail of the NRT and reprocessed OLCI datasets and the reprocessed multi sensor datasets for the Baltic Sea the geographical box [53.3, 65.8] degrees north and [9.25, 30.3] degrees east (orange areas in Figure 1). A brief scientific and technical description of the products and of the procedures used to retrieve them is provided below.

II.3 Production temporal coverage

The MY archive for the OLCI datasets is regularly extended with the MY interim production.

II.4 Production chain

This section describes the processing chain for the data production in the Baltic Sea of near real-time (NRT) and reprocessed time series (MY) of surface chlorophyll concentration (CHL) and of the Phytoplankton Functional Types (PFT) datasets derived from OLCI and multi sensor reflectances. CHL is retrieved by an ensemble scheme to combine Chl-a obtained with different MLP algorithms developed for the Baltic Sea and applied to OLCI and OC-CCI remote sensing reflectances R_{rs} after a band-shifting to the Sea-viewing Wide Field-of-view Sensor (SeaWIFs) spectral bands.

II.4.1 Level-3 (L3) remote-sensing reflectances

Primary input data for the retrieval of the biogeochemical variables are the merged Sentinel 3A and 3B OLCI remote-sensing reflectances (Rrs) at 300m spatial resolution and the Level-3 (L3) remote-sensing reflectances (Rrs) at ~ 1 km spatial resolution derived from daily L3 the ESA-CCI (Climate Change Initiative).

II.4.1.1 OLCI remote-sensing reflectances

The assessment of the OLCI remote-sensing reflectances for the Baltic CDOM-dominated waters have shown poor performances for the OLCI R_{rs} retrieved with the EUMETSAT (European Organisation for the Exploitation of Meteorological Satellites) Operational Baseline (Zibordi et al., 2018; Zibordi et al., 2022) and the alternative atmospheric correction processing chain (Cazzaniga et al., 2021). Hence, the OLCI NRT and MY processing chain presented in this document relies on Sentinel-3/OLCI Level1 EUMETSAT operational data production and a dedicated atmospheric correction from OLCI Level-1 to Level-2 remote-sensing reflectances is carried out for the OLCI sensors on both Sentinel 3A and 3B. Based on a round-robin of atmospheric correction algorithm based on a comparison with *in situ* radiometry, POLYMER V4.14 (Steinmetz et al., 2011; Steinmetz and Ramon, 2018) was selected as it never retrieved negative reflectance values in the 400-490 nm spectral range (Gonzalez-Vilas et al., 2024; Figure 2).

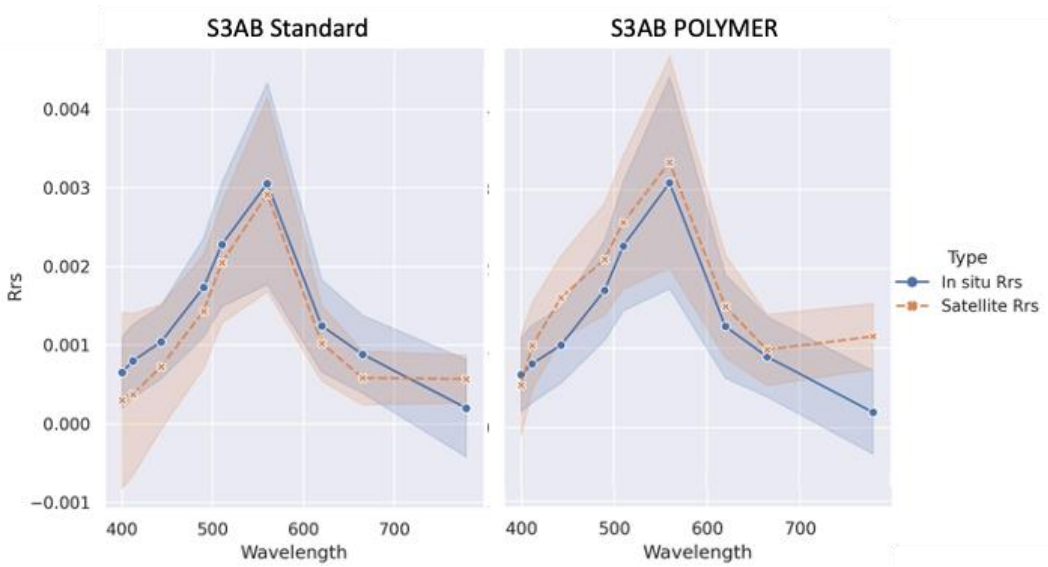


Figure 2: Comparison with in situ radiometry of the OLCI R_{rs} retrieved with the EUMETSAT V3.0.1 standard processor and the POLYMER V4.14 atmospheric correction (Gonzalez-Vilas et al., 2024).

The POLYMER remote-sensing reflectances for Sentinel 3A and 3B are remapped at the sensor nominal spatial resolution on an equi-rectangular grid with the "nearest value" interpolation using the "pyresample" python package. Then, the R_{rs} fields from two sensors are merged without any bias correction, as for both sensors the System Vicarious Calibration gains has been implemented by EUMETSAT to adjust to the Top-of-Atmosphere absolute radiance levels.

II.5 Multi-sensor remote-sensing reflectances

The multi-sensor remote-sensing reflectances for the reprocessed products are available at six MERIS wavebands (Medium Resolution Imaging Spectrometer 412-443-490-510-560-665 nm), as the entire data time series is processed with a consistent configuration, providing users with the longest and most consistent available data time series, spanning over twenty years (1997 to 2023), for the Baltic Sea. Within Copernicus Marine Service, the spectral R_{rs} was produced by the Plymouth Marine Laboratory (PML), using the OC-CCI processor version 6.0[3] (to merge at 1km resolution (rather than at 4km for the global product OCEANCOLOUR_GLO_BGC_L3_MY_009_107) MERIS, MODIS-AQUA (Moderate Resolution Imaging Spectroradiometer on AQUA satellite), SeaWiFS, VIIRS (Visible Infrared Imaging Radiometer Suite on NPP and NOAA20 satellites) and OLCI (Ocean and Land Colour Instrument on Sentinel3A and Sentinel3B satellite) data.

For the OC-CCI multi-sensor production, atmospheric correction is performed independently for each sensor, and the merging is performed for the calibrated reflectances, after proper band shift to the MERIS bands (Mélin and Sclep, 2015; Sathyendranath et al 2019). The dataset used in this version, hereafter referred to as OC-CCIv6, incorporates the NASA (National Aeronautics and Space Administration) R2018.0 reprocessing for MODIS-AQUA, SeaWiFS and VIIRS, the ESA 4th reprocessing for MERIS and EUMETSAT v3.0.1 reprocessing for OLCI A and B. All sensors apart from SeaWiFS were processed to remote-sensing reflectances with POLYMER (Steinmetz et al., 2011; Steinmetz and Ramon, 2018).

II.6 Chlorophyll algorithm

This section presents the evolution of the ensemble approach for the Chl-a retrieval in the Baltic Sea. The background scheme to combine different regional algorithms (Brando et al. 2021) is reviewed in Sec. II.6.1 and updated in Sec. II.7 to enhance the quality of results. Elements underlying operational applications are then considered in Sec. II.7.1, together with a summary performance analysis.

II.6.1 Background ensemble scheme

The regional Multi-Layer Perceptron (MLP) neural network to derive Chl-a estimates in the Baltic Sea was originally developed based on field measurements collected within the BiOMaP program of JRC/EC (Joint Research Centre of the European Commission; Zibordi et al., 2011, D'Alimonte et al., 2011; D'Alimonte et al., 2014). The motivation to use the MLP for the *Chl-a* retrieval is that this algorithm permits exploiting R_{rs} input values at individual bands, which can convey additional information with respect to the sole spectral slope of the band-ratio algorithms. The MLP, however, lacks extrapolation capabilities, and this requires careful verification of the validity range.

The optical properties of the Baltic Sea are so complex that they demand to draw on all technical and scientific resources at hand. By seeking performance enhancements, Brando et al. (2021) have shown how the Chl-a retrieval can be improved by merging different MLPs rather than using individual ones.

[3] www.esa-oceancolour-cci.org)

To this end, they introduced the concept of *ensemble Chl-a estimate* (Chl-a_{ENS3}), adopting a solution similar to the so-called "mixture of experts" in statistical pattern recognition (Yuksel et al 2012). The innovative idea was to rely on a set of MLPs that use R_{rs} at different wavelengths as input, rather than a single MLP based on all wavelengths (or a fixed subset of wavelengths) only. Each MLP can have a different relevance due to the spectral heterogeneity of the uncertainty induced by the atmospheric correction process on R_{rs} values.

The MLPs merging was set through the concept of novelty of the input R_{rs} spectrum within the BIOMAP data adopted for the MLP development. The comparison of results obtained from R_{rs} at different bands have shown that the most reliable ensemble is based on the combination of the three MLPs reported in Table 4. In practice, the data novelty served to balance the contribution from different algorithms in a weighted summation, allowing for an improved Chl-a retrieval. The downside, however, is that this way of merging represents a heuristic solution rather than a genuine statistical approach.

Table 4: Multi-Layer Perceptron (MLP) neural network pool adopted for the ENS ensemble scheme.

Index (i)	Training data set	Function type	Acronym	Input bands (nm)
1	BIOMAP	Multi-Layer Perceptron	MLP_{3b}	490, 510 and 555
2	BIOMAP	Multi-Layer Perceptron	MLP_{4b}	490, 510, 555 and 670
3	BIOMAP	Multi-Layer Perceptron	MLP_{5b}	443, 490, 510, 555 and 670

The review of results and performance assessments with extended validation data sets have then addressed the analysis toward new methodological developments. It was specifically found that, for a better Chl-a retrieval in the Baltic Sea, the background ensemble approach should be revisited to embrace the following additional properties:

1) be fully grounded on probability estimates,
2) provide an out-of-range (OOR) flag, and
3) be consistent with other class membership definitions adopted in OC (e.g., seawater optical types; Moore et al. 2009, 2014; Mélin and Vantrepotte 2015).

As detailed in the next section, the background ensemble scheme is here reformulated following these guidelines.

II.7 Regional chlorophyll algorithm: generalized ensemble scheme to combine Chl-a obtained with different MLPs

The mathematical expressions for the MLP implementation are presented below, followed by a description of the ensemble method revision into a new formulation based on the Cumulative Distribution Function (CDF) (Figure 3).

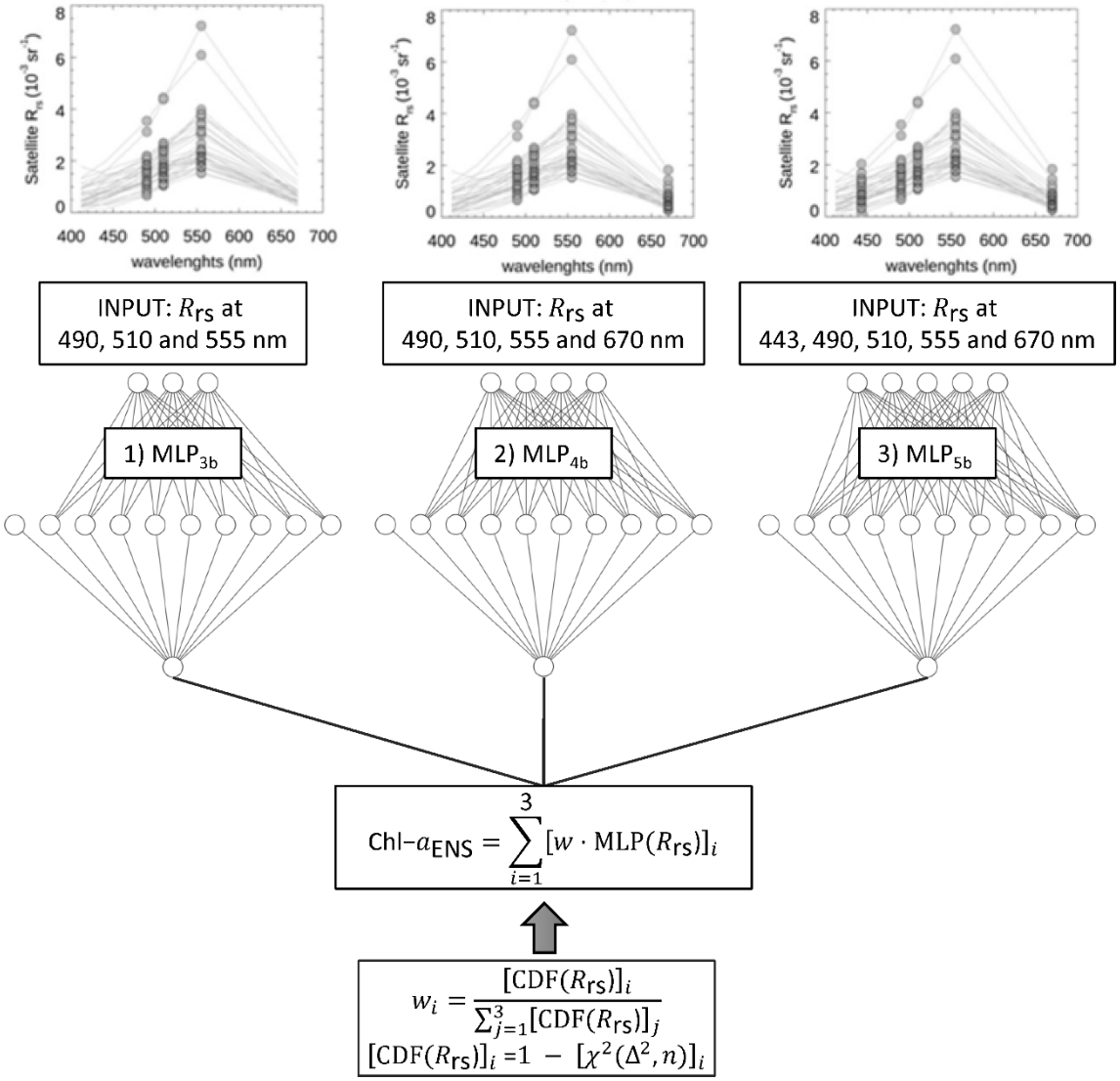


Figure 3: Overview of the ensemble scheme to combine Chl-a obtained with different Multi-Layer Perceptron (MLP) neural netwoks for the Baltic Sea, as an evolution of the background approach presented by Brando et al. (2021)—see also Table 4.

The Multi-layer Perceptron scheme

The MLP scheme consists of successive layers of units, with each unit in one layer connected to each unit of the next layer. The input R_{rs} values at the selected spectral wavelengths are firstly log-transformed **l** = $\log_{10}(R_{rs})$, and then z-score scaled x = (l - μ_l) / σ_l, where μ_l and σ_l indicate respectively the mean and standard deviation of log-transformed R_{rs} of the BiOMaP training data (D'Alimonte et al., 2011; D'Alimonte et al., 2014).

Let y=*f*(x) denote the MLP function, where the input vector x has entries x_i for *i*=1,…,d. The value of the hidden unit *j*, denoted a_j, is a linear combination of the input quantities:

$$a_j = \mathrm{b}_j^{(1)} + \sum_{i=1}^{d} \mathrm{w}_{ji}^{(1)} x_i \tag{1}$$

where $\mathrm{w}_{ji}^{(1)}$ represent the weights linking the input unit i to the hidden unit j, and $\mathrm{b}^{(1)}_j$ is the bias adaptive parameter. The activation of the hidden unit j, identified with z_j and representing the input for the next layer, is obtained as:

$$z_j = g(a_j) = \frac{e^{a_j} - e^{-a_j}}{e^{a_j} + e^{-a_j}}, \tag{2}$$

with *g* indicating the hyperbolic tangent activation function. The output *y* is computed as:

$$y = \mathrm{b}^{(2)} + \sum_{i=1}^{\mathrm{M}} \mathrm{w}_j^{(2)} z_j, \tag{3}$$

where M indicates the number of hidden units with weight w^2_j and $\mathrm{b}^{(2)}$ is the bias coefficient.

Then, data post-processing converts the *y* value into the final result as $\mathrm{MLP} = 10^{(y \cdot \sigma_c + \mu_c)}$, where μ_c and μ_c are the mean and the standard deviation of log-transformed Chl-*a in situ* measurements, respectively.

The CDF ensemble scheme

Each MLP of the ensemble pool is based on a different set of R_{rs} input bands (Table 4). The ith member of the pool is denoted as $[\mathrm{MLP}(R_{\mathrm{rs}})]_i$ in the following equations (an equivalent notation is adopted for the related quantities, for instance, by indicating the corresponding input remote sensing reflectance as $[R_{\mathrm{rs}}]_i$).

The updated MLPs ensemble (see Figure 3) is computed as

$$\text{Chl-}a_{\mathrm{ENS3_CDF}} = \sum_{i=1}^{3} [w \cdot \mathrm{MLP}(R_{\mathrm{rs}})]_i \tag{4}$$

estimating the weight factor

$$[w]_i = \frac{[\mathrm{CDF}(R_{\mathrm{rs}})]_i}{\sum_{j=1}^{N} [\mathrm{CDF}(R_{\mathrm{rs}})]_j}. \tag{5}$$

based on the following CDF value

$$[\mathrm{CDF}(R_{\mathrm{rs}})]_i = 1 - [\chi^2(\Delta^2, n)]_i, \tag{6}$$

where

- the number of degrees of freedom *n* of the Chi-squared distribution function χ^2 is equal to the number of R_{rs} input bands, and
- Δ^2 is the squared Mahalanobis distance

$$\Delta^2 = (\boldsymbol{x} - \boldsymbol{\mu})\boldsymbol{\Sigma}(\boldsymbol{x} - \boldsymbol{\mu}) \tag{7}$$

of the log-transformed input water reflectance spectrum $\boldsymbol{x} = \log(R_{rs})$, with $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ representing the mean and the covariance of log-transformed training data, respectively, at the same band set of R_{rs} (D'Alimonte et al. 2003).

As the MLPs used in this retrieval scheme were developed at SeaWIFs spectral bands (D'Alimonte et al., 2011; Brando et al., 2021), prior to the application of the CHL retrieval, the R_{rs} values for the OLCI and OC-CCI spectral bands are band-shifted to the SeaWIFs spectral bands by means of inverse and direct application of the Quasi Analytical Algorithm (QAA; Mélin and Sclep, 2015; Lee et al., 2009). The QAA was modified to warrant a non-negative phytoplankton absorption at any band, to avoid malfunctioning of the band-shift procedure in this absorption dominated waters (Brando et al., 2021).

This CDF definition corresponds to the probability that the training set includes a data point with a Mahalanobis distance that is larger than that of the input spectrum $[R_{rs}]_i$ (i.e., $[\chi^2(\Delta^2, n)]_i$)—see Figure 4. Hence, the CDF is one when the point is the center of the data distribution. It is important to notice that the CDF value does not depend on the number of bands (Brereton 2015, Etherington 2019). Consequently, the weight *w* of an input remote sensing reflectance that is statistically consistent with the training data is, in principle, the same regardless of which bands have been selected.

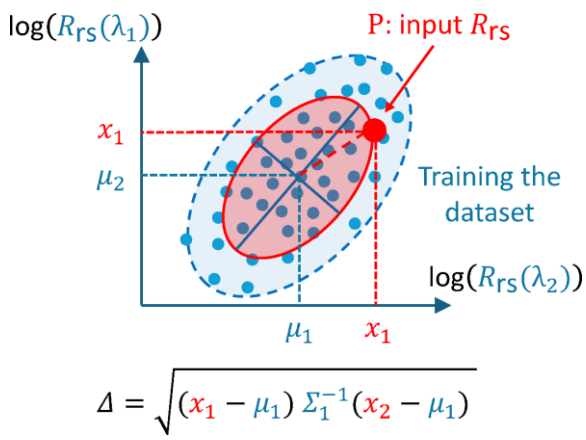


Figure 4: Schematic of the Mahalanobis distance employed to compute the Multi-Layer Perceptron (MLP) neural network weight for the ensemble.

The new blending formulation relies on the same Mahalanobis distance and the Chi-squared distribution function already adopted for the definition of Optical Water Type (OWT) (Moore et al. 2009, 2014; Mélin and Vantrepotte 2015). This consistency allows establishing a correspondence between the regional algorithms ensemble and the optical water variability, although the former drops the requirement of preserving the R_{rs} bands set between classes and can hence be viewed as a generalization of the latter.

There is also some analogy between the newly formulated blending framework and the hypothesis testing method (e.g., validity flag based on a CDF threshold versus the confidence level). The main difference, however, is that the hypothesis testing assesses statistical parameters of the data distribution; instead, the CDF is here applied for inference over a single input data point (which implies that terms such as "P-value" and "confidence level" should not be used in this context). For this reason, the new method of combining different regional MLPs is denoted as *CDF ensemble*.

II.7.1 Elements underlying operational applications of the CDF ensemble for the Chl-a retrieval in the Baltic Sea

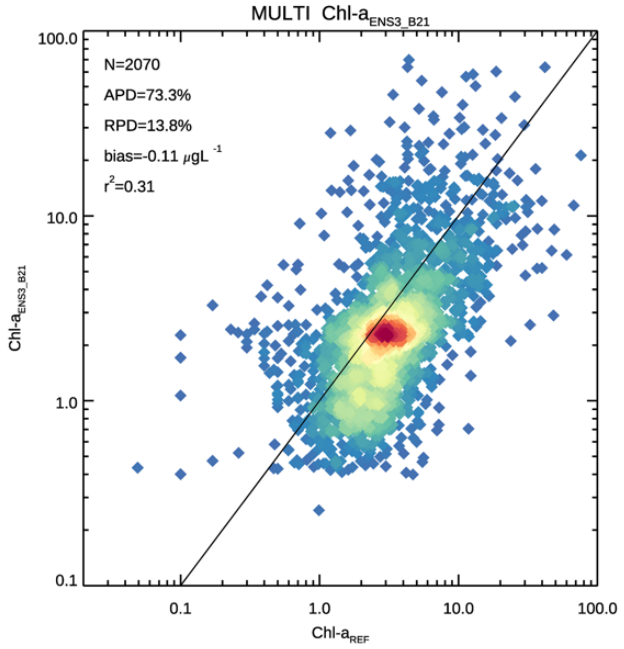
The performance of the CDF ensemble (Chl-$a_{\mathrm{ENS3_CDF}}$) has been evaluated by adopting the following scheme to facilitate operational applications:

1) The merging is done with the MLPs for which the input R_{rs} at the selected band subset has a CDF above a threshold = 0.001.
2) If the criterion 1) is not satisfied, then the Chl-a derived only with the MLP3b (i.e., the most stable one as a result of manifold tests) is provided as a *ghost* solution flagged as out-of-range (OOR).

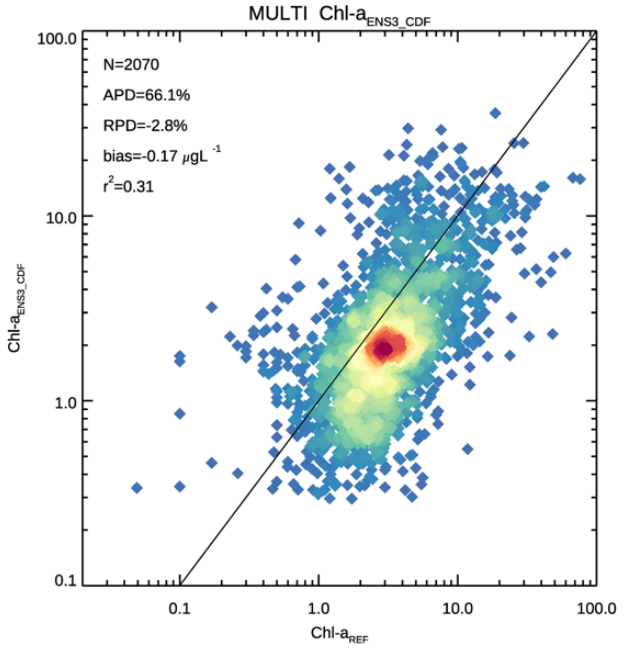
The capability of the CDF ensemble to outperform the background scheme is documented in Figure 5 by considering the matchups for the Multi-sensor and OLCI datasets.

By checking the agreement between the R_{rs} input to the MLP and the BIOMAP data set, the CDF ensemble reduces the probability of deriving Chl-a values from incompatible remote sensing reflectances, whatever the underlying cause of differences. Note that this concept can be extended to any functional form of the regional algorithms and other training data (which is behind the scope of the present analysis). It is additionally highlighted that the statistical compatibility between the input R_{rs} and MLP training data is a necessary but not sufficient condition for the validity of results. In fact, an input water reflectance with a spectral shape not included in the BIOMAP dataset could be transformed into a compatible one by chance (e.g., due to an atmospheric correction artifact). This latter case unavoidably contributes to the uncertainty in the Chl-a retrieved with the present CDF ensemble formulation.

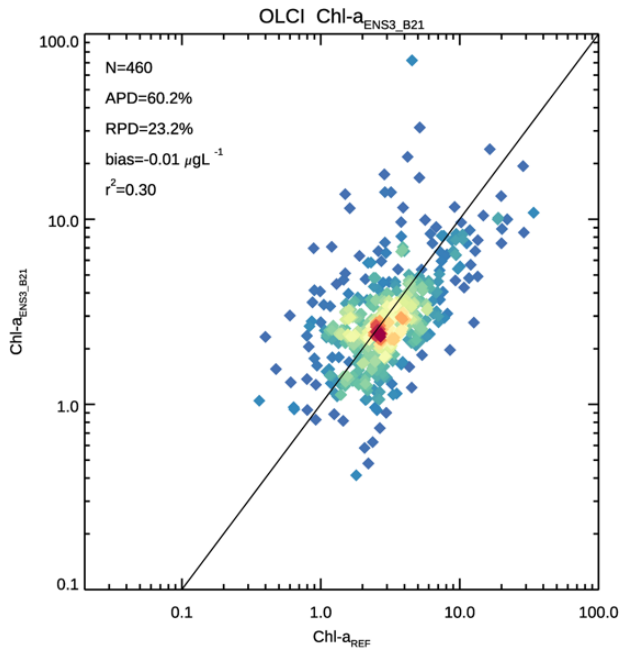
In conclusion, the CDF ensemble inherits the advantages offered by the background scheme based on the novelty detection, providing an adequate way to overcome some of its limitations, such as the impossibility of defining a single novelty threshold that has the same statistical meaning when considering the input water reflectance a different number of input bands. Additional elements underpinning operational applications are the OOR definition and a *modus operandi* correspondence with classification methods for the OWT definition. The performance assessment based on matchup data has confirmed that the generalized framework helps to improve Chl-a estimates, but also with a remark on the intrinsic performance limitation represented by any R_{rs} spectra that appear compatible with the training data due to artifacts.



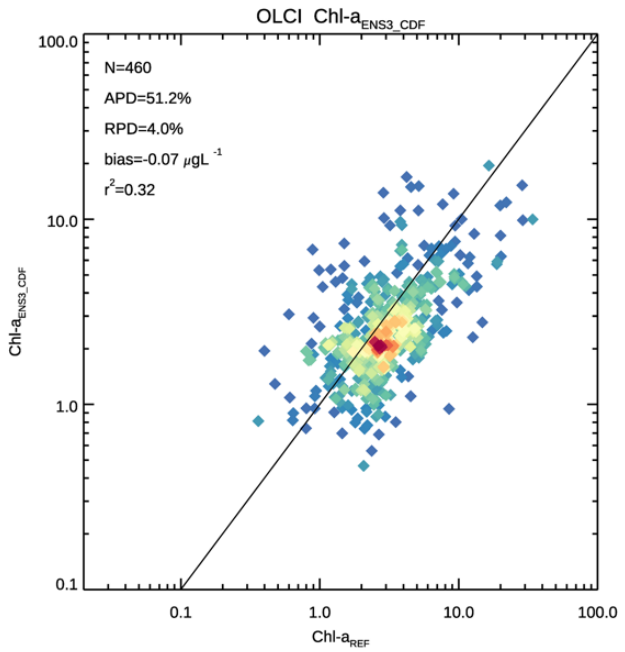
(a)



(b)



(c)



(d)

Figure 5: Assessments of the background (Chl-$\boldsymbol{a}_{ENS3_B21}$) (a); and the new CDF ensemble (Chl-$\boldsymbol{a}_{ENS3_CDF}$) (b) based on the MULTI-SENSOR matchup data. Same assessments with the OLCI matchup dataset in (c) and (d) (see text for details).

II.7.2 PFT/PSC algorithm

PFT product is computed applying regional algorithms developed for the Baltic Sea (Di Cicco et al., in preparation) to the BAL MY CHL dataset. These algorithms are empirical relationships developed following a global approach (Hirata et al., 2011), also regionalized for the Mediterranean Sea in Di Cicco et al. (2017), based on the relation between phytoplankton abundance and the trophic status of the environment (Brewin et al., 2010, 2011; Chisholm, 1992; e.g., Devred et al., 2006; Uitz et al., 2006). The functions results from the existing covariability between *in situ* PFT groups (expressed as chlorophyll concentration) and the corresponding log10-transformed *in situ* total chlorophyll a (TChla) concentration, taking account of the log-normal distribution of this pigment (Chisholm, 1992; Hirata et al., 2011). The in-situ pigmental dataset used for the PFT algorithm development in the Baltic Sea are fully detailed in Section III.1.

PFT in-situ quantification

Following previous works carried out for global and regional applications (Barlow et al., 1993; Gieskes et al., 1988; Uitz et al., 2006; Di Cicco et al., 2017; Meler et al., 2020), the contribution of each phytoplankton group to the total chlorophyll a of the Baltic assemblage was computed performing a multiple regression analysis between *in situ* TChla and seven pigments considered diagnostic (DP) for the main algal groups (Claustre, 1994; Vidussi et al., 2001), to provide the best estimates of the TChla–DP (algal groups) ratios specific for the Baltic Sea (see Di Cicco et al., 2017 for more details). Our analysis highlights that two of the seven typical DP (19'-butanoiloxyfucoxanthin and 19'-hexanoiloxyfucoxanthin, diagnostic mainly for coccolithophores) do not show a good correlation with TChla, probably due to their very low concentration or total absence in the most of the samples (as also found in Meler et al., 2020). A stepwise method was applied to verify the effect of these two variables on the regression model. The results, in terms of Root Mean Squared Error (RMSD) and determination coefficient (r^2) between the TChla estimated from the model and in-situ TChla, highlights that the inclusion of these two parameters doesn't make significant improvements in the model (same r^2 and RMSD excluding and including the two pigments respectively). Based on these considerations, these DP were excluded by the regression model (as in Meler et al., 2020) and, accordingly, here Nano component does not include Haptophytes group. Table 5 shows the final selected pigments with their taxonomic meaning and the best estimates resulting from the multiple regression analysis with their standard deviation and significance level.

Table 5: Final selected pigments with their taxonomic group and the best estimates resulting from the in-situ regression analysis. Standard deviation and significance level of each coefficient are also shown.

DP (& Taxonomic meaning)	Coefficients	Standard Deviation	p-Value
Fucoxanthin (Diatoms)	2.77	0.06	< 0.001
Peridinin (Dinoflagellates)	2.91	0.13	< 0.001
Alloxanthin (Cryptophytes)	3.73	0.09	< 0.001
Total Chlorophyll b (Green algae & Prochlorophytes)	1.27	0.07	< 0.001
Zeaxanthin (Prokaryotes)	1.82	0.1	< 0.001

Each coefficient, multiplied by the concentration of the corresponding DP, gives the chlorophyll concentration associated with the PFT Diato, Dino, Crypto, Green and Prokar (as defined in section I.1).

It's important to point out that DP analysis of our dataset highlights that divinil-chlorophyll a and b are not quantified for any samples (so could be below the detection threshold or not present in the samples). Thus, the Green group for the Baltic Sea consist only of Green Algae. It should also be noted that Zeaxanthin, widely used as a marker pigment for cyanobacteria, in the Baltic Sea is mainly found in small single-cell *Picocyanobacteria* species, including *Synechococcus* spp. which play a significant role during the Baltic summer blooms (Mazur-Marzec et al. 2013). Wojtasiewicz et al., (2016) detected Zeaxanthin also in some filamentous cyanobacteria of Baltic Sea, in particular in *Apahanizomenon flos-aquae*, typical of the Baltic summer blooms. Nevertheless, Zeaxanthin seems lack in several filamentous cyanobacteria, like for example *N. Spumigena* and *Anabaena* sp., that also occur in the Baltic Sea with an important role in the summer cyanobacterial blooms, and mainly detected by phycobilins (Wojtasiewicz et al., 2016). Unfortunately, we have no availability of a such complete dataset with High Performance Liquid Chromatography (HPLC) pigments and additional phycoblins, also covering the basin at this spatial and temporal scale. Thus, for the taxonomic meaning of the group it is necessary to keep these considerations in mind.

With regards to Micro, Nano and Pico, as well as in Di Cicco et al., (2017), the identification and quantification of these size classes is based on the known relation between taxonomic groups and their typical dimensions (defined in Sieburth et al., 1978), following the approach of Vidussi et al. (2001). In addition, following the suggestion of Chase et al. (2020) and Meler et al (2020), the latter properly related to the Baltic Sea, for the definition of the size group we assign the contribution of the Green group in equal parts to Pico and Nano. Indeed, in this work micro (20–250 µm) is defined as Diato and Dino, nano (2–20 µm) includes Crypto and half of the Green group and Pico (0.2–2 µm) is referred to Prokar and half Green.

PFT Model development

The in-situ dataset of PFT-Chl concentration resulted from the DP analysis for the Baltic Sea was divided in two independent subsets: 70% and 30% of the data (646 samples), respectively for the algorithm calibration and validation. Due to a limited number of matchups between our in-situ dataset and the available satellite chlorophyll data required to validate also the satellite application of the new models, the dataset has been divided so that: all the in-situ data not useful for matchup analysis (60% of all in-situ dataset) has been used for the calibration of the new empirical algorithms; the remaining 40% in-situ data, available instead also for matchups analysis, has been randomly divided in order to obtain an independent validation subset (30% of all in-situ data) to evaluate the performance of the new models, and in order to enlarge the calibration dataset, increasing it to 70% of all available data.

To identify the empirical algorithm for the PFT, a regression analysis between the in-situ Chl concentration related to each group and the corresponding in-situ log10-transformed total Chl concentrations (considering the log-normal distribution of this pigment) was performed, considering the known co-variability existent between them (Chisholm, 1992; Hirata et al., 2011). The regression analysis was performed applying the Nonlinear Least Squares method with a robust fit by minimizing the least absolute residuals (LAR) and using the Trust-region algorithm for the fitting procedure. The analysis resulted in four empirical exponential relationships for Pico, Nano, Dino and Green while Micro, Diato, Crypto and Prokar has been derived as difference to maintain the mass balance with the chlorophyll. Figure 6 shows the final algorithms. Their coefficients and functional forms are provided in Table 6. The range of applicability of the algorithms is defined between 0.13 to 25.5 mg m-3 of Chl concentration.

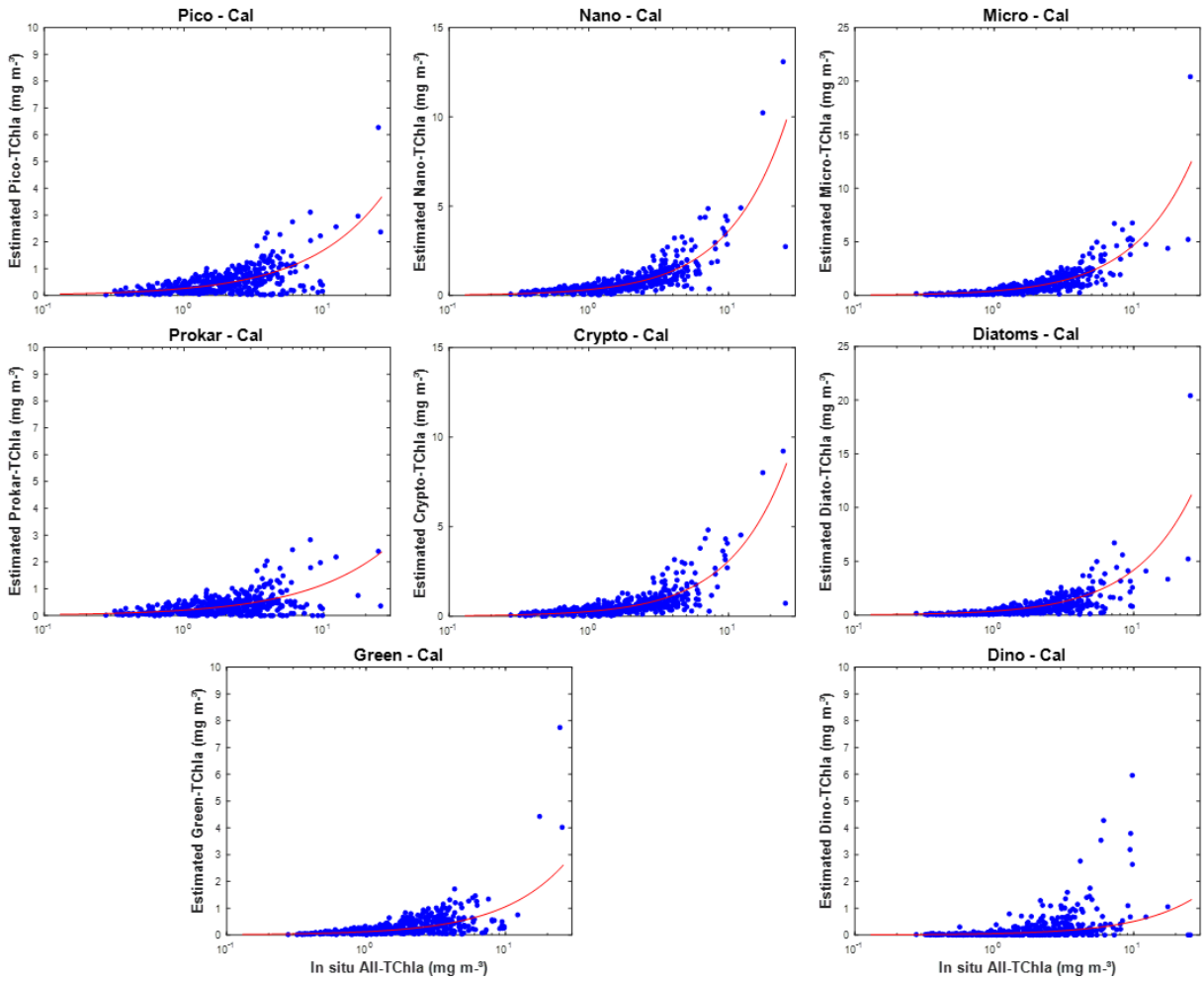


Figure 6: Regional relationship (in red) between in-situ TChla and TChla related to each phytoplankton class (Pico, Nano, Micro, Prokar, , Crypto, Diato, Green, Dino) for the Baltic Sea.

Table 6: Coefficients and functional forms of the algorithms developed for the PFT estimates in the Baltic Sea.

PFTs	**Function**	**a**	**b**
PICO	$a * \exp(b * x)$	0.261	1.870
NANO	$a * \exp(b * x)$	0.324	2.412
MICRO	$Chl - Pico - Nano$	–	–
DINO	$a * \exp(b * x)$	0.050	2.313
DIATO	$Micro - Dino$	–	–
GREEN	$a * \exp(b * x)$	0.119	2.181
PROKAR	$Pico - 0.5 * Green$	–	–
CRYPTO	$Nano - 0.5 * Green$	–	–
			$with:\ x = log_{10}(Chl)$

The hindcast evaluation of the algorithm performance (i.e. the same calibration dataset was used for fitting and testing) and their metrics are shown in section IV.3 (Table 13), where the validation of the algorithms with an independent in-situ dataset is also fully described.

II.7.3 Cyanobacterial subsurface and surface summer blooms

Moreover, consistent with the HELCOM environmental reporting (e.g. Öberg 2017) the cyanobacterial subsurface and surface summer blooms are detected by applying the thresholds on R_{rs} at the wavelength of 555 and 670 nm (Hansson and Håkansson 2007). The thresholds applied to daily R_{rs} images were set at $R_{rs}(555)$ > 4.25 10^{-3} sr^{-1} and $R_{rs}(670)$ > 1.22 10^{-3} sr^{-1} for the subsurface and surface blooms, respectively, as defined in Kahru et al (2007). This R_{rs}-based approach to detect cyanobacterial blooms allows overcoming the difficulty of retrieving accurate $Chl\text{-}a$ values in summer bloom conditions (Hansson and Håkansson 2007; Brando et al 2021).
As cyanobacterial blooms are extremely patchy and the surface scum is unlikely to totally cover a 1 km resolution pixel , it is possible for both thresholds to be exceeded concurrently. Hence, the cyanobacteria summer bloom mask variable (CYANOBLOOM) is the sum of the values for two bloom types (subsurface =1; surface=2) thus each pixel could be a value between 0 (no summer bloom detected) and 3 (both subsurface and surface summer blooms detected).

III VALIDATION FRAMEWORK

For these NRT and MY product, the Consiglio Nazionale delle Ricerche (CNR) performs only the offline validation. The online validation of the OLCI NRT product against the long-term climatology will be resumed once the climatology with a spatial resolution of 300m is computed with this processing chain.

III.1 Offline Validation

Offline validation refers to the estimate of the statistical parameters listed in Table 7 and Table 8, based on the pairwise comparison of the satellite estimation $(X^E)_{i=1,\dots,N}$ to the *in situ* observations $(X^M)_{i=1,\dots,N}$ (whose space-time distribution is shown in Figure 7).

The Baltic Sea is a focal point of activities for protection of the marine environment and is inter-governmentally organized by the Helsinki Commission, HELCOM. In that framework various data are gathered by contracting parties in the COMBINE database, which is hosted by the International Council for the Exploration of the Sea (ICES). Amongst others, the database includes chlorophyll-a measurements from the Baltic Sea. This dataset gives a long-time overview of chlorophyll concentrations from the 1970ies to 2021. For this work, the ICES dataset was augmented with the *in situ* CHL dataset assembled by SYKE based on Alg@line data collected on the Helsinki -Travemünde, Helsinki – Stockholm and Kemi - Travemünde routes from 1997 to 2023. Water samples collected with sequence water sampler from the flow-through water (nominal water inlet 5 m) were filtered with glass fibre filters (Whatman GF/F); chlorophyll-a was extracted with ethanol and then chlorophyll content was determined fluorometrically (Kaitala et al., 2008). Satellite CHL was extracted as the median of the 3x3 pixels box around the respective *in situ* match-up collected between 9-15 local time, (i.e. 10-16 UTC), for a minimum of five valid pixels.

In-situ Rrs data are Level-2 normalized water-leaving radiances, corrected for bi-directional effects and referred to nadir, taken from automatic and quality-controlled measurements at three AERONET-OC sites (Zibordi et al, 2009; Zibordi et al., 2021): Gustav Dalen Tower (58.594 N, 17.467 E), Helsinki Lighthouse (59.949 N, 24.925 E) and Irbe Lighthouse (57.751N, 21.723E). These data were transformed to Rrs and band-shifted to the OLCI and OC-CCI spectral bands (Mélin and Sclep, 2015) by means of inverse and direct application of the QAA algorithm (Lee et al., 2009). The QAA was modified to warrant a non-negative phytoplankton absorption at any band, to avoid malfunctioning of the band-shift procedure in this absorbtion dominated waters (Brando et al., 2021). Output data are provided a few times a day. Eligible match-ups were chosen as the median of all spectra within two hours of the local noon for the OC-CCI datasets and within two hours of the Sentinel 3 overpasses for the OLCI datasets. Match-ups belong to the period 2016-2021 for OLCI and 2005-2021 for OC-CCI.

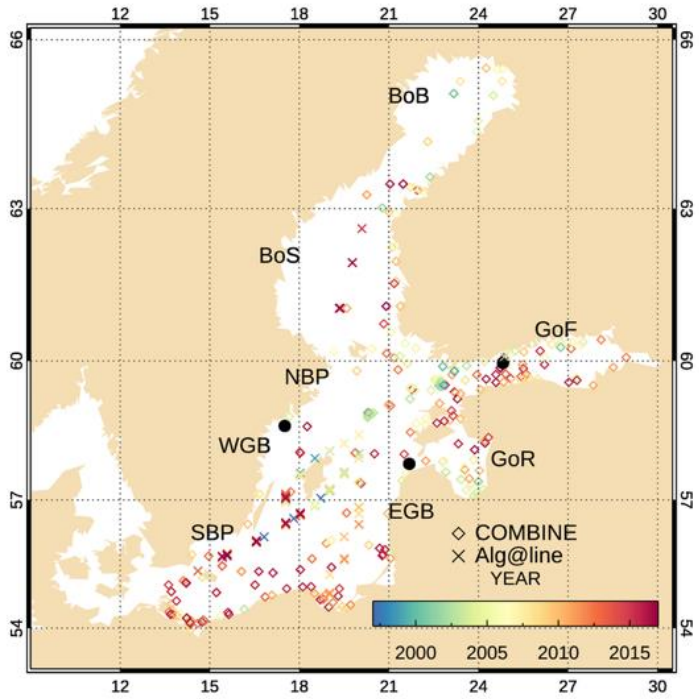


Figure 7: Location of the match-ups in the in situ dataset used for the offline validation: crosses identify the in situ Alg@line data by SYKE; the diamonds are the COMBINE data extracted from the ICES database. The Black dots mark the location of the three AERONET-OC sites.

High Performance Liquid Chromatography (HPLC) pigment data required for the model development and for the offline validation come from the analytical laboratories of the Joint Research Centre (JRC) and the Institute of Oceanology of the Polish Academy of Sciences (IOPAN), thanks to the effort of dr. Elisabetta Canuti and dr. Joanna Stoń-Egiert, respectively (Canuti et al., 2016; Stoń-Egiert et al., 2005). JRC phytoplankton samples were collected in six different cruises from 2004 to 2008 in the framework of Bio-Optical mapping of Marine Properties (BiOMaP, Zibordi et al., 2011) programs, carried out by the JRC and covering almost the entire basin. The IOPAN sampling activities have been conducted nearly every month from 2009 to 2018, focusing mainly on the South Baltic Region of the Bornholm Basin, covering the open waters of the Polish economic zone and the Gulf of Gdansk region. IOPAN activities were supported in the framework of Statutory Research Projects of the Institute and projects funded by the European Union (Satellite Monitoring of the Baltic Sea Environment, SatBałtyk and Oceanographic Data and Information System[4]). Considering that the natural samples were collected by different teams and analyzed in different laboratories, quality assurance analyses based on Aiken et al. (2009) were performed, to ensure the consistency of the two datasets. The final useful dataset consists of 646 stations, with TChla values ranging from 0.13 to about 25.5 mg m−3. Data are shown in Figure 8 (more details in Di Cicco et al., in preparation). The dataset was divided in two independent subsets, 70% and 30% of all data, for the algorithm calibration and validation respectively (as detailed in section II.7.2).

[4] https://ecudo.pl/

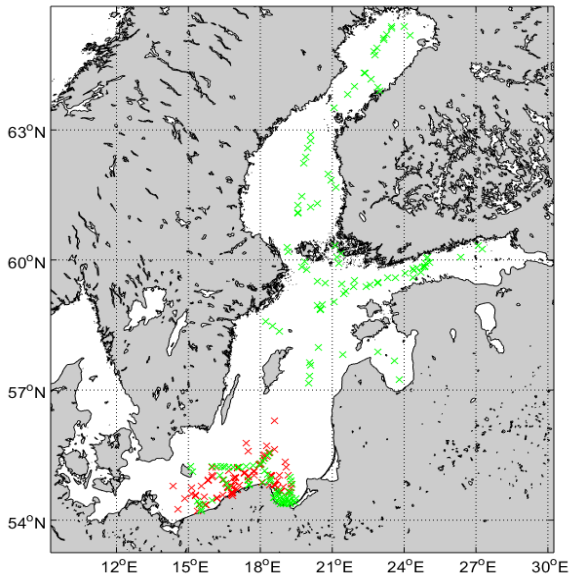


Figure 8: HPLC pigment dataset used for the development of the regional PFT algorithms (70% of all) and for the in-situ and satellite validation (30% of all). Green crosses identify the JRC subset and red crosses IOPAN one. Baltic Sea.

Satellite CHL for the PFT matchup analysis was extracted as the median of the 3x3 pixels box around the respective *in situ* match-up, for a minimum of four valid pixels. The total number of match-ups was 194.

Table 7: List of metrics used to compare the estimated (satellite-based) dataset $X^E_{i,i=1..N}$ *to a reference (in situ) dataset* $X^M_{i,i=1..N}$ *.*

Name	Definition
Estimated dataset mean (X^E)	$\overline{X}^E = \frac{1}{N}\sum_{i=1}^{N} X_i^E$
Reference dataset mean (X^M)	$\overline{X}^M = \frac{1}{N}\sum_{i=1}^{N} X_i^M$
Type-2 slope *(S)*	$S = \frac{\sum_{i=1}^{N}\left(X_i^E-\overline{X}^E\right)^2 - \sum_{i=1}^{N}\left(X_i^M-\overline{X}^M\right)^2 + \left[\left\{\sum_{i=1}^{N}\left(X_i^E-\overline{X}^E\right)^2 - \sum_{i=1}^{N}\left(X_i^M-\overline{X}^M\right)^2\right\}^2 + 4\left\{\sum_{i=k}^{N}\left(X_k^E-\overline{X}^E\right)\left(X_k^M-\overline{X}^M\right)\right\}^2\right]^{1/2}}{\sum_{i=k}^{N}\left(X_k^E-\overline{X}^E\right)\left(X_k^M-\overline{X}^M\right)}$
Type-2 intercept *(I)*	$I = \overline{X}^E - S\cdot\overline{X}^M$
Determination coefficient *(r^2)*	$r^2 = \frac{\left[\sum_{i=1}^{N}\left(X_i^E-\overline{X}_i^E\right)\left(X_i^M-\overline{X}_i^M\right)\right]^2}{\sum_{i=1}^{N}\left(X_i^E-\overline{X}_i^E\right)^2\sum_{i=1}^{N}\left(X_i^M-\overline{X}_i^M\right)^2}$

Table 7: (continued) List of metrics used to compare the estimated (satellite-based) dataset $X^{E}_{i,i=1..N}$ *to a reference (in situ) dataset* $X^{M}_{i,i=1..N}$ *.*

Names	Definitions
Root Mean Square Difference *(RMSD)*	$RMSD = \sqrt{\frac{\sum_{i=1}^{N}\left(X_i^E - X_i^M\right)^2}{N}}$
Centre-pattern Root Mean Square Difference *(cRMSD)*	$cRMSD = \sqrt{\frac{\sum_{i=1}^{N}\left\{\left[X_i^E - \left(\sum_{j=1}^{N} X_j^E\right)\right] - \left[X_i^M - \left(\sum_{k=1}^{N} X_k^M\right)\right]\right\}^2}{N}}$
Bias	$bias = \frac{1}{N}\sum_{i=1}^{N}(X_i^E - X_i^M)$
Mean Relative Percentage Difference (RPD)	$RPD = \frac{1}{N}\sum_{i=1}^{N}\left(\frac{X_i^E - X_i^M}{X_i^M}\right) x\ 100$
Mean Absolute Percentage Difference (APD)	$APD = \frac{1}{N}\sum_{i=1}^{N}\left\lvert\frac{X_i^E - X_i^M}{X_i^M}\right\rvert x\ 100$
Mean Absolute Difference (MAD)	$MAD = \frac{1}{N}\sum_{i=1}^{N}\left\lvert X_i^E - X_i^M\right\rvert$

Notes:

- For log-normally distributed variables (such as Chl) both datasets are log-transformed prior to computing the metrics.
- Type-2 regression (called also orthogonal regression) is used instead of minimising the vertical distance between independent data and linear fit (as in Type-1 regression), which minimises the perpendicular distance between independent data and linear fit. Type-1 regression typically assumes the dependent variable (*in situ* data) is known perfectly well, when in reality, the *in situ* data are also affected by uncertainties (e.g. problems with *in situ* data sampling techniques) that are difficult to quantify. Therefore,
- A slope (S) close to one and an intercept (I) close to zero is an indication that the model compares well with the *in situ* data.
- The Centre-pattern (or unbiased) Root Mean Square Difference describes the error of the estimated values with respect to the measured ones, regardless of the average bias between the two distributions.
- The RMSD is thus the distance, on average, of a data point from the fitted line, measured perpendicular to the regression line. The RMSD is directly interpretable in terms of measurement units, and so is a better measure of goodness of fit than a correlation coefficient.
- The bias indicator is also directly interpretable in terms of measurement units. The squaring is not done so negative values can cancel positive values. The bias is not a good indicator of

average model performance and might be a misleading indicator of average error. It should be carefully considered taking into account the scatter plot.

- The Mean Relative Percentage Difference (RPD) and the Mean Absolute Percentage Difference (APD) are always computed in linear space, also for log-normally distributed variables.

Table 8: Summary of metrics. For the product description see section I.1.

Variable	Supporting observation	Class	Metrics
CHL	*in situ* data from international data bases	4	See Table 3
Rrsxxx	AERONET-OC towers	4	See Table 3

IV VALIDATION RESULTS

IV.1Remote Sensing Reflectance

IV.1.1 NRT and MY OLCI datasets

The L3 OLCI R_{rs} are compared with the *in situ* AERONET-OC Rrs (Figure 9). The validation statistics associated with the analysis, the Estimated Accuracy Numbers, are shown in Table 9.

The density scatter plot of the OLCI R_{rs} vs *in situ* automated radiometric measurements (Figure 9) shows that atmospheric correction performed adequately in such optically complex waters in terms of correlation for the 443-665 nm spectral range (Determination coefficient 0.62-0.89) and relative and absolute percentage differences for the 490-665 nm spectral range (APD and RPD: 11-30%). The 400, 412 and 779 nm bands show higher uncertainties (Determination coefficient of 0.36, 0.52 and 0.25, APD and RPD). Nevertheless, a positive bias is observed across the whole spectral range (bias ranging 1-6 10^{-4} sr^{-1}).

The 400, 412 and 779 nm spectral bands are not used as input for the retrieval of any of the derived products, so the higher uncertainties can be deemed acceptable.

The number of matchups varies across the spectral bands (Table 9) as AERONET-OC instruments explicitly designed to match OLCI spectral bands were introduced gradually in the Baltic Sea sites from 2016 onwards.

Table 9: Estimated Accuracy Numbers for the NRT/MY OLCI Rrs as defined in section III.1. Estimated dataset mean (X^E); Reference dataset mean (X^M); Type-2 slope (S); Type-2 intercept (I); Coefficient of determination (r^2); Root Mean Square Difference (RMSD); Centre-pattern Root Mean Square Difference (cRMSD); Bias; Mean Relative Percentage Difference (RPD); Mean Absolute Percentage Difference (APD).

Variable	X^M	X^E	S	I	r^2	RMSD	cRMSD	Bias	RPD	APD	N
RRS400	6.46E-4	6.42E-4	1.808	-5.26E-4	0.365	5.12E-4	5.12E-4	-4.03E-6	22.7	143.9	557
RRS412	8.05E-4	1.13E-3	1.244	1.26E-4	0.516	5.05E-4	3.89E-4	3.22E-4	77.1	87.2	646
RRS443	1.03E-3	1.67E-3	1.227	4.07E-4	0.658	7.10E-4	3.05E-4	6.40E-4	75.2	75.5	646
RRS490	1.72E-3	2.16E-3	1.192	1.06E-4	0.747	5.58E-4	3.47E-4	4.37E-4	27.9	29.1	646
RRS510	2.25E-3	2.60E-3	1.097	1.32E-4	0.809	4.99E-4	3.55E-4	3.50E-4	17.3	18.9	557
RRS560	3.07E-3	3.39E-3	1.052	1.62E-4	0.896	5.34E-4	4.26E-4	3.22E-4	11.9	14.0	646
RRS620	1.24E-3	1.53E-3	1.101	1.60E-4	0.783	4.09E-4	2.93E-4	2.85E-4	26.8	28.7	557
RRS665	8.73E-4	9.83E-4	1.007	1.04E-4	0.714	2.70E-4	2.46E-4	1.10E-4	23.1	29.8	646
RRS779	1.61E-4	1.14E-3	3.172	6.26E-4	0.254	1.04E-3	3.44E-4	9.76E-4	1087.3	1727.6	557

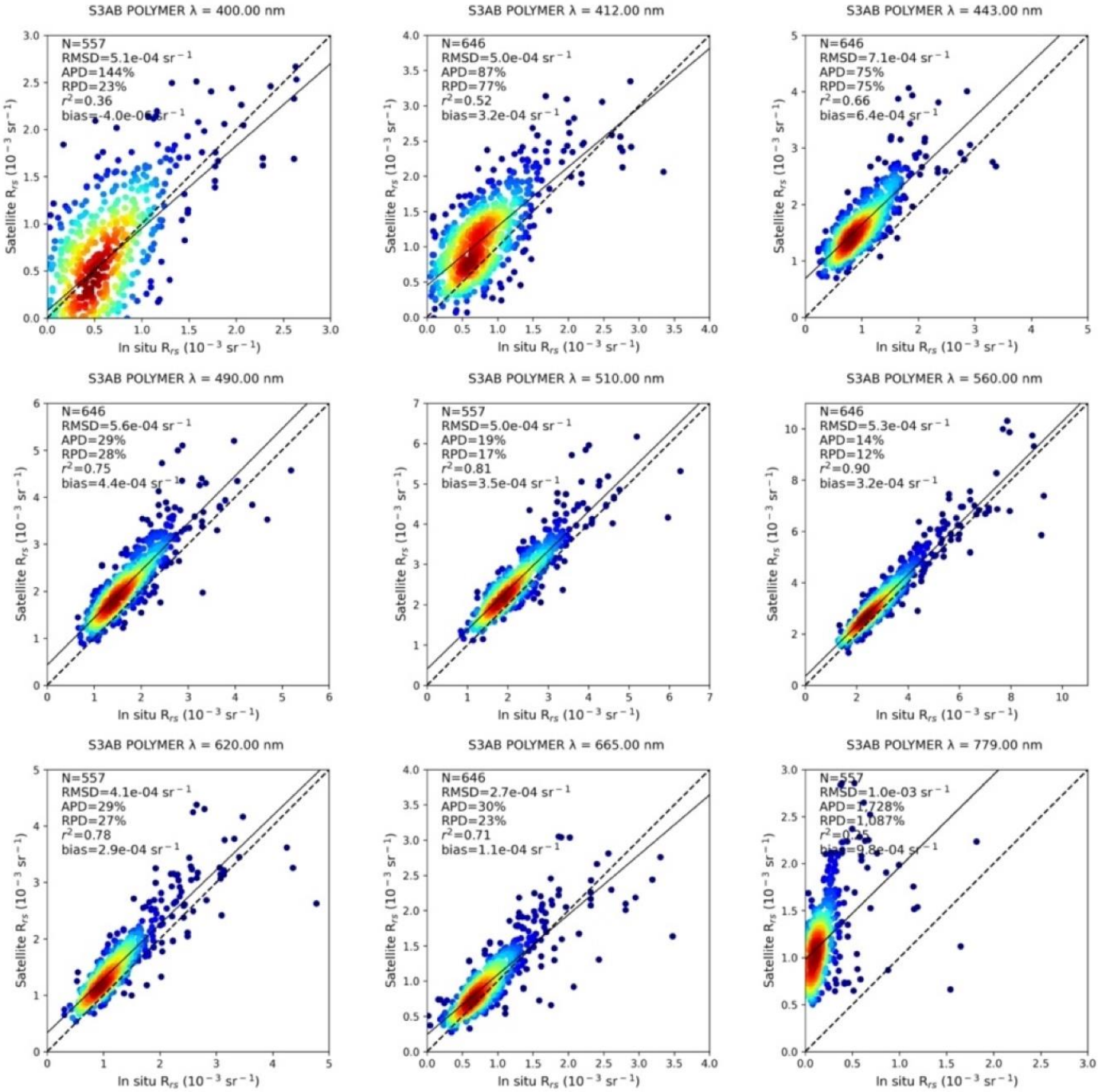


Figure 9: Scatter plot of the NRT/MY OLCI satellite-derived and in situ Rrs space-time co-located observations over the Baltic (BAL). Data points are colour-coded by density.

IV.1.2 MY multi-sensor datasets

The density scatter plot of the multi-sensor CCIv6.0 Rrs vs. *in situ* (Figure 10) shows that atmospheric correction performed adequately in such optically complex waters in terms of correlation for the 442-665 nm spectral range (Determination coefficient 0.80-0.90,

Table 10), while the 412 and 442 nm bands show higher uncertainties (Determination coefficient of 0.42 and 0.62). Nevertheless, a positive bias is notable for the 412-510 nm spectral range (bias ranging 2-6 10^{-5} sr^{-1}), with only the 560 and 665 nm were overall centred on the 1:1 line with a low bias of ~6 10^{-5} sr^{-1}.

The overall similar behaviour across the two classes of datasets is due to same atmospheric correction being applied. In facts, in OC-CCIv6 all sensors apart from SeaWiFS were processed to remote-sensing reflectance with POLYMER.

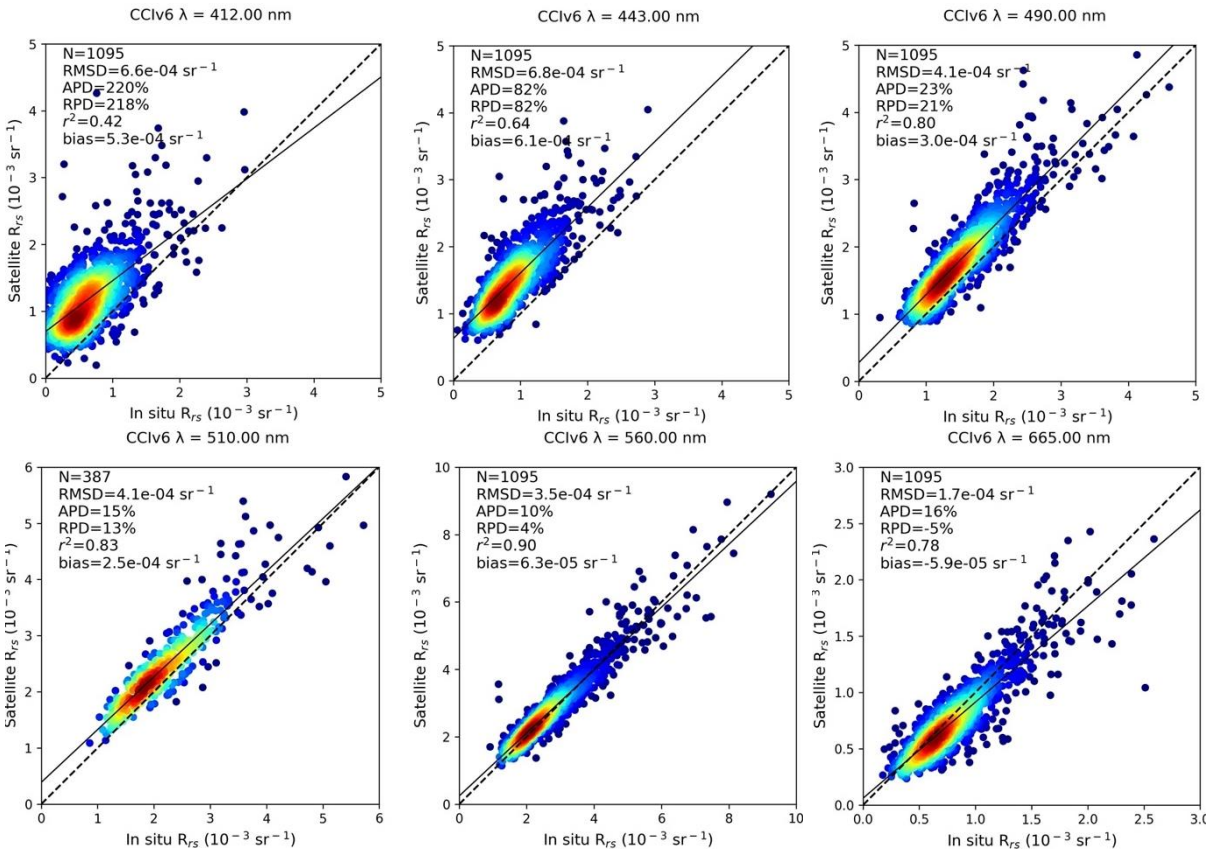


Figure 10: Scatter plot of the MY (reprocessed) multi-sensor satellite-derived and in situ Rrs space-time co-located observations over BAL. Data points are color-coded by density (red: highest density; blue: lowest density).

Table 10: Estimated Accuracy Numbers for Rrs as defined in section III.1. Estimated dataset mean (X^E); Reference dataset mean (X^M); Type-2 slope (S); Type-2 intercept (I); Coefficient of determination (r^2); Root Mean Square Difference (RMSD); Centre-pattern Root Mean Square Difference (cRMSD); Bias; Mean Relative Percentage Difference (RPD); Mean Absolute Percentage Difference (APD).

Variable	X^M	X^E	S	I	r^2	RMSD	cRMSD	Bias	RPD	APD	N
RRS412	6.82E-4	1.22E-3	1.291	3.35E-4	0.416	6.62E-4	3.91E-4	5.34E-4	217.8	220.0	1095
RRS443	9.16E-4	1.53E-3	1.283	3.51E-4	0.642	6.78E-4	2.95E-4	6.10E-4	81.6	81.7	1095
RRS490	1.55E-3	1.85E-3	1.151	6.09E-5	0.797	4.07E-4	2.80E-4	2.96E-4	21.1	22.9	1095
RRS510	2.26E-3	2.51E-3	1.037	1.68E-4	0.827	4.13E-4	3.29E-4	2.50E-4	12.8	15.2	387
RRS560	2.73E-3	2.79E-3	0.982	1.12E-4	0.902	3.51E-4	3.46E-4	6.28E-5	3.7	9.7	1095
RRS665	8.12E-4	7.53E-4	0.961	-2.74E-5	0.781	1.74E-4	1.64E-4	-5.86E-5	-5.5	16.3	1095

IV.2CHL

Figure 11 presents the scatter plot of the satellite-derived and *in situ* CHL co-located observations over the Baltic for the three element ensemble (Chl-a_{ENS3_CDF}) applied to OC-CCI and OLCI remote sensing reflectances R_{rs} after a band-shifting to the SeaWIFs spectral bands. The Estimated Accuracy Numbers associated with the analysis are reported in Table 11.

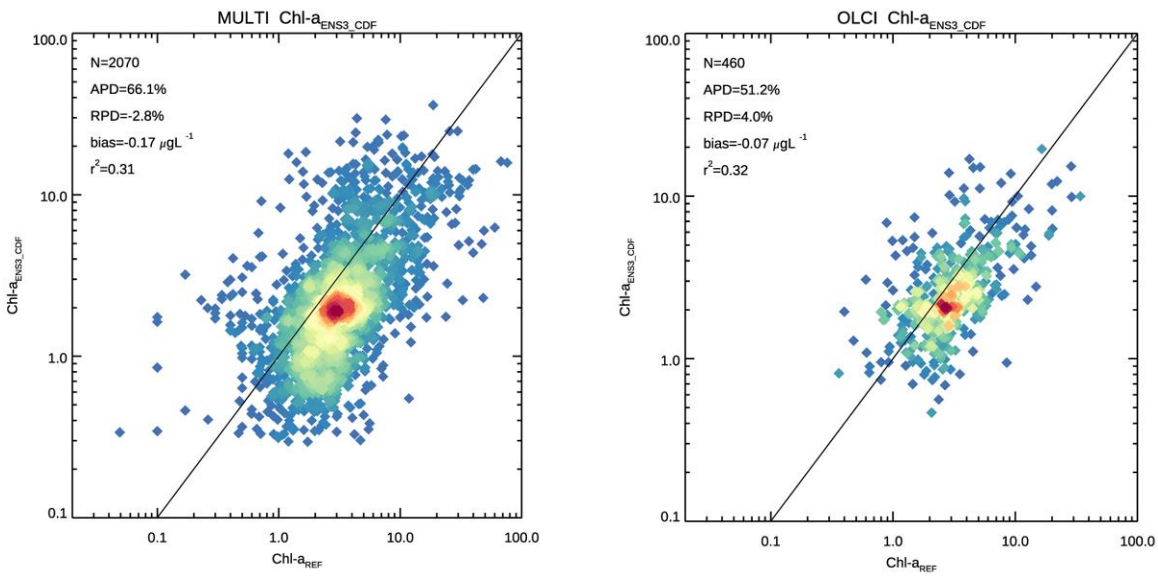
The performance of the new Ensemble MLP Chlorophyll algorithm (Chl-a_{ENS3_CDF}) show a Determination coefficient of 0.312 and RPD and APD of -2.8 and 66.1% for the MY multi sensor datasets (Figure 11, Table 11). These results are an improvement from the previous version of the ensemble (Chl-a_{ENS3}) that yielded a Determination coefficient of 0.312 and RPD and APD of 13.8 and 73.3% (Table 11, Figure 5). The new version shows better performance in terms of RPD and APD, but a slightly larger negative bias. Overall the effect of the positive bias observed for multi-sensor CCIv6.0 Rrs in the 412-510 nm spectral range (Figure 10,
Table 10) seems to be adequately handled by both 3 element ensemble solutions.

The comparison for the NRT ad MY OLCI 300m Chl-a_{ENS3_CDF} datasets yielded a Determination coefficient of 0.324 and RPD and APD of 4 and 51.2% (Figure 11, Table 11). Also for OLCI this was an improvement in comparison with the previous version of the ensemble (Chl-a_{ENS3}) that yielded a Determination coefficient of 0.324 and RPD and APD of 4 and 60.2% (Table 11, Figure 5).

Some of the differences in the statistics for the 1 Km MY multi sensor and the OLCI 300m datasets are likely due the different uncertainties of the Rrs input spectra. Nevertheless, the EANs for the OLCI results are consistent with the multi sensor datasets even if there was a lower number of matchups available from 2016 to date (460 vs 2070).

Table 11: Estimated Accuracy Numbers for CHL retrieved with previous (Chl-α_{ENS3_B21}) and the new CDF ensemble (Chl-α_{ENS3_CDF}) applied to OLCI and OC-CCI Rrs. Estimated dataset mean (X^E); Reference dataset mean (X^M); Type-2 slope (S); Type-2 intercept (I); Coefficient of determination (r^2); Root Mean Square Difference (RMSD); Centre-pattern Root Mean Square Difference (cRMSD); Bias; Mean Relative Percentage Difference (RPD); Mean Absolute Percentage Difference (APD).

CHL	X^M	X^E	S	I	r^2	RMSD	cRMSD	Bias	N	RPD	APD	MAD
MULTI Chl-α_{ENS3_B21}	0.491	0.380	1.132	-0.176	0.312	0.356	0.339	-0.111	2070	13.8	73.3	2.827
MULTI Chl-α_{ENS3_CDF}	0.491	0.323	1.093	-0.214	0.312	0.375	0.335	-0.168	2070	-2.8	66.1	2.568
OLCI Chl-α_{ENS3_B21}	0.474	0.462	0.864	0.053	0.302	0.271	0.271	-0.012	460	23.2	60.2	1.968
OLCI Chl-α_{ENS3_CDF}	0.474	0.403	0.826	0.012	0.324	0.271	0.262	-0.071	460	4.0	51.2	1.833



(a) (b)

Figure 11: Scatter plots of the satellite-derived and in situ CHL co-located observations over the Baltic (BAL) with the new ensemble (Chl-a_{ENS3_CDF}) applied to OC-CCI and OLCI R_{rs}.. Data points are colour-coded by density (red: highest density; blue: lowest density).

IV.3PFT/PSC

A qualification of the Baltic Sea PFT dataset is provided together with the offline validation. This latter was performed using an independent in-situ validation dataset (described in section II.7.2) and by applying the new algorithms to the BAL MY CHL dataset. To evaluate the consistency of these results, the matchup analysis was compared with the hindcast evaluation of the models (i.e. the same calibration dataset was used for fitting and testing) and with the in-situ validation analysis performed

applying the new algorithms to the in-situ chlorophyll data from the same independent validation dataset. The comparison is shown in Figure 12 (for Micro, Nano and Pico) and Figure 13 (for the remaining groups). The metrics are summarized in Table 12 (offline validation) and Table 13 (hindcast evaluation and validation of the algorithms).

Figure 12, Figure 13 and Table 13 show a very good agreement between the in-situ validation and hindcast evaluation. The results of satellite validation (Table 12) are consistent with the metrics of the in-situ validation in particular for the Micro, Pico, Diato and Prokar groups. For those groups with a wider chlorophyll dynamic range (i.e. Nano, Dino, Crypto and Green), the statistics partially deteriorates when the models were applied to satellite CHL. In these groups, fits show the best performances when the algorithms were applied to intermediate values of satellite CHL. The data points deviate from the 1:1 line at CHL values lower of 0.3/0.5 mg m-3 and greater than 8/10 mg m-3 following the behavior of the satellite Chl. The main reason of this behavior is probably related to a fewer number of in-situ data available for the validation in this range of concentrations, combined with a dispersion increase also detectable in the validation of the MY BAL Chl concentration for these *in situ* values. This is because, since the abundance algorithms are based on chlorophyll concentration, their performance is obviously related also to the performance of chlorophyll retrieval.

However, this behavior is expected considering the complex optical characteristics of the Baltic Sea, and despite these considerations, in general, the metrics indicate a quite good predictive power for the models, also consistent with the one of chlorophyll.

Table 12: Estimated Accuracy Numbers for Phytoplankton Functional Types (PFT) as defined in section III.1. Estimated dataset mean (X^E); Reference dataset mean (X^M); Type-2 slope (S); Type-2 intercept (I); Coefficient of determination (r^2); Root Mean Square Difference (RMSD); Centre-pattern Root Mean Square Difference (cRMSD); Bias; Mean Relative Percentage Difference (RPD); Mean Absolute Percentage Difference (APD).

Variable	X^M	X^E	S	I	r^2	RMSD	cRMSD	Bias	RPD	APD	MAD	N
Micro	0.528	0.614	0.565	-0.055	0.413	0.326	0.319	0.065	55.3	84.7	0.250	192
Nano	0.446	0.474	0.277	-0.227	0.179	0.490	0.489	0.027	109.3	153.9	0.369	191
Pico	0.509	0.352	0.401	-0.336	0.238	0.350	0.311	-0.160	-6.5	58.3	0.281	192
Diato	0.425	0.544	0.621	-0.034	0.389	0.326	0.308	0.107	72.2	96.8	0.250	190
Dino	0.104	0.081	0.239	-0.858	0.162	0.516	0.503	-0.111	34.2	93.4	0.389	145
Crypto	0.446	0.390	0.285	-0.308	0.178	0.493	0.490	-0.058	72.1	127.8	0.375	191
Green	0.177	0.168	0.299	-0.549	0.145	0.415	0.415	-0.023	43.8	93.2	0.341	191
Prokar	0.315	0.270	0.421	-0.358	0.281	0.299	0.291	-0.068	7.7	54.5	0.234	182

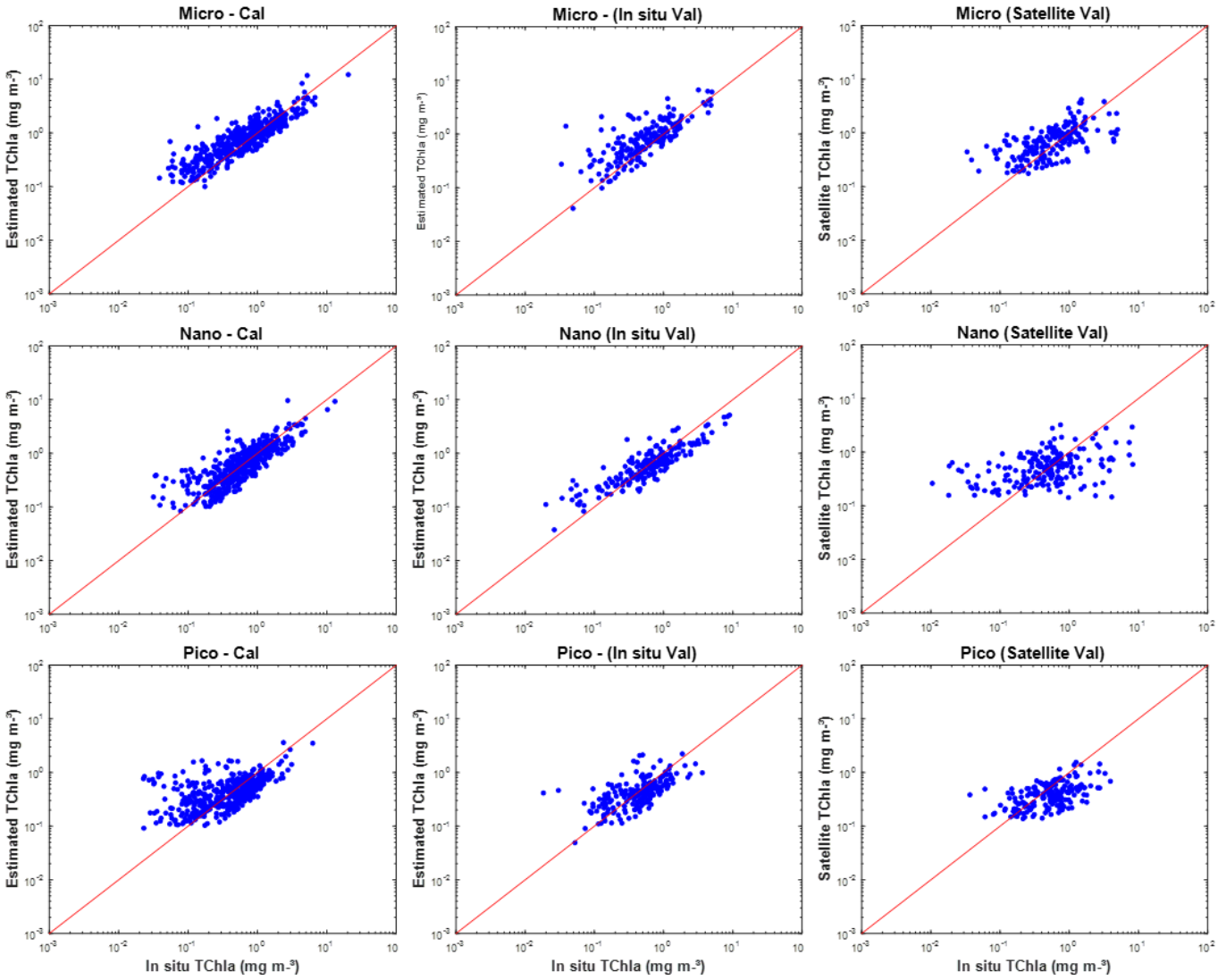


Figure 12: Comparison of the offline validation of the Phytoplankton Functional Types (PFT) dataset (right column) with the hindcast evaluation of the new algorithms (left column) and the in-situ validation (central column) for Micro, Nano and Pico groups. In situ and satellite validation were regressed with in-situ observation obtained from an independent validation dataset (see section II.7.2).

Table 13: Statistical results of the in situ analysis: hindcast evaluation (new Phytoplankton Functional Types -PFT- regional algorithms were applied to the in situ Chl of the calibration dataset, 70% of all data) and in-situ algorithm validation (new PFT regional algorithms were applied to the Chl of the independent validation dataset, 30% of all data).

Variable	r^2	RMSD	Bias	RPD	APD	MAD	N
Micro – Cal	0.783	0.247	0.103	48	64	0.190	452
Micro – Val	0.578	0.315	0.157	94	105	0.217	194
Nano – Cal	0.646	0.239	0.019	26	55	0.179	452
Nano – Val	0.768	0.246	0.010	23	56	0.194	194
Pico – Cal	0.291	0.370	0.077	103	136	0.263	452
Pico – Val	0.391	0.286	0.014	39	75	0.218	194
Diato – Cal	0.654	0.339	0.183	98	112	0.262	452
Diato – Val	0.498	0.348	0.194	103	115	0.258	192
Dino – Cal	0.456	0.419	-0.101	20	77	0.332	373
Dino – Val	0.496	0.393	0.006	42	86	0.313	147
Crypto – Cal	0.533	0.303	0.055	57	87	0.224	450
Crypto – Val	0.725	0.298	0.035	42	76	0.235	193
Green – Cal	0.443	0.325	0.019	45	85	0.261	451
Green – Val	0.459	0.328	0.062	55	87	0.263	193
Prokar – Cal	0.343	0.335	0.063	62	96	0.267	420
Prokar – Val	0.311	0.292	-0.001	28	66	0.228	184

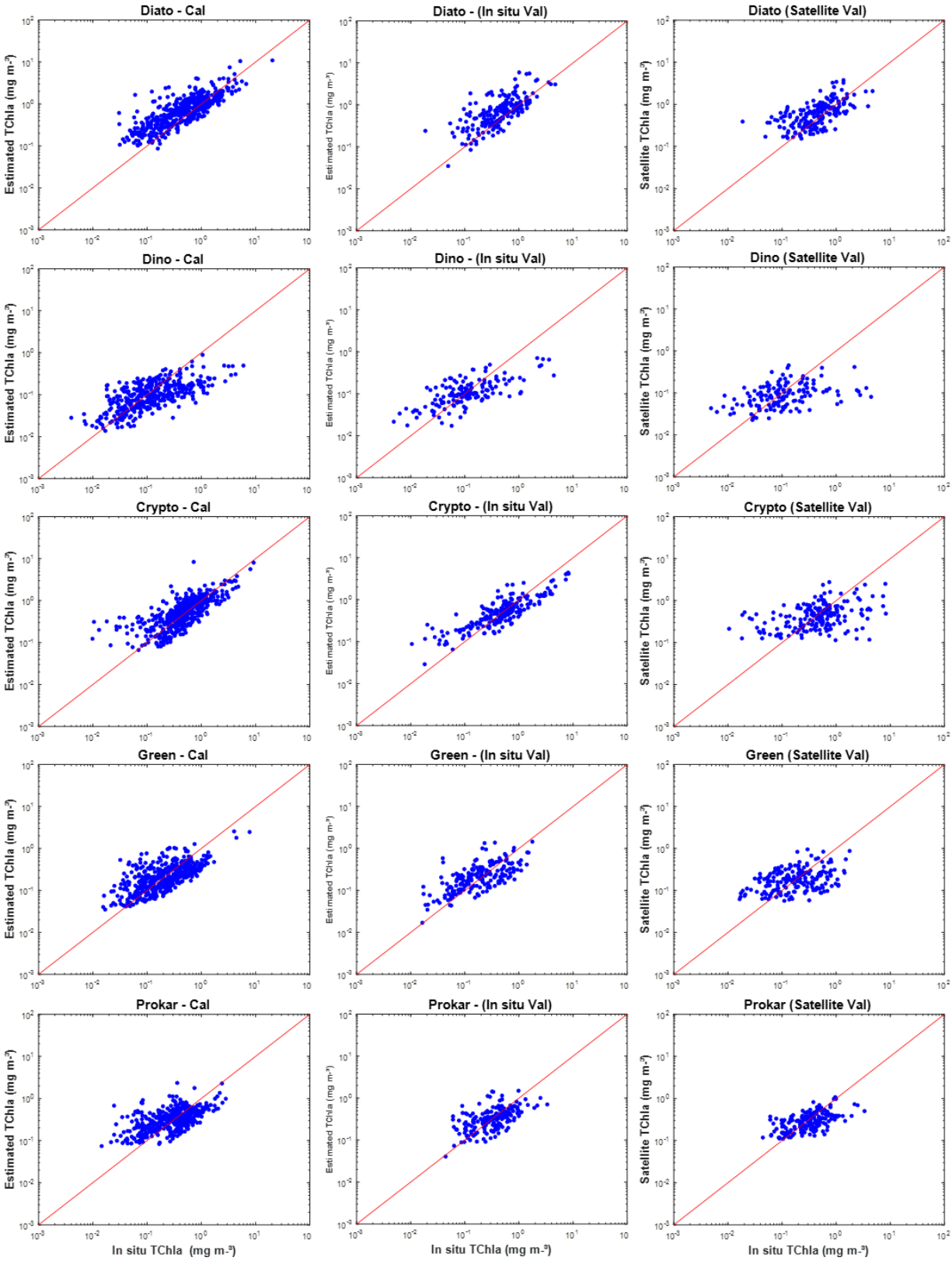


Figure 13: Comparison of the offline validation of the Phytoplankton Functional Types (PFT) dataset (right panels) with the hindcast evaluation of the new algorithms (left panels) and the in-situ validation (central panels) for Diato, Dino, Crypto, Green and Prokar groups. In situ and satellite validation were regressed with in-situ observation obtained from an independent validation dataset (see section II.7.2.

V SYSTEM'S NOTICEABLE EVENTS, OUTAGES OR CHANGES

This section is dedicated to describe and track changes that may occur in the current operational system including Include changes due to outages (data gaps), upstream data (e.g., addition or loss of a sensor, inclusion/removal of in-situ platforms), reprocessing, corrections, assimilated data, boundary conditions, among others, which may impact product quality (see Table below).

Issue	Changes
November 2024	This version presented the qualification of the OLCI NRT and fully reprocessed MY time series (2016-2023) and the fully reprocessed MY time series (1997-2023) based on CCIv6.0. The POLYMER atmospheric correction scheme is used to retrieve remote-sensing reflectances for the Sentinel3A+B OLCI 300m datasets. The multi-sensor 1 km datasets are based on CCIv6.0 that incorporates the NASA R2018.0 reprocessing for MODIS-AQUA, SeaWiFS and VIIRS, the ESA 4th reprocessing for MERIS and EUMETSAT v3.0.1 reprocessing for OLCI A and B. All sensors apart from SeaWiFS were processed to remote-sensing reflectances with POLYMER. The CHL and PFT datasets in the NRT and MY products are retrieved with a consistent processing chain using the updated 3 elements ensemble algorithm (Chl-a_{ENS3_CDF}).
November 2022	This version presented the qualification of the OLCI NRT and fully reprocessed MY time series (2016-2021). A new atmospheric correction scheme was introduced for the Sentinel3A+B OLCI merged datasets, replacing the BAL HZG OLCI processing chain that was retired. This version presented the qualification of the fully reprocessed MY time series (1997-2021) based on CCIv6.0 and incorporates the NASA R2018.0 reprocessing for MODIS-AQUA, SeaWiFS and VIIRS, the ESA 4th reprocessing for MERIS and EUMETSAT v3.0.1 reprocessing for OLCI A and B. All sensors apart from SeaWiFS were processed to remote-sensing reflectances with POLYMER. The CHL and PFT datasets are now retrieved with a consistent processing chain for the NRT and MY products using the 3 elements ensemble algorithm (Chl-a_{ENS3}) introduced in 2021 for the MY products.
July 2022 (For MY) CMEMS-OC-QUID-009-133to134	The July-2022 release only concerned adoption of a new CMEMS nomenclature for products and datasets. Naming of the product and datasets is changed to adopt general naming rules of CMEMS. It includes some changes concerning auxiliary variables (flags) and some individual files are grouped together (e.g. all RRS in the same file). 009_133 supersedes 009_080 & 009_087 that are retired at EIS. This version reported the qualification of the fully reprocessed the time series (1997-2020) based on CCIv4.2 that incorporates NASA atmospheric correction and the R2018.0 reprocessing for MODIS-AQUA, SeaWiFS and VIIRS, and POLYMER atmospheric correction for MERIS R2012.0.
July 2022 (for OLCI NRT) CMEMS-OC-QUID-009-131to132	The July-2022 release only concerned adoption of a new CMEMS nomenclature for the OLCI products and datasets. Naming of the product and datasets is changed to adopt general naming rules of CMEMS. It includes some changes concerning auxiliary variables (flags) and some individual files are grouped together (e.g. all RRS in the same file). 009_131 supersedes 009_048 & 009_049 that are retired at EIS. 009_132 supersedes 009_332 that is retired at EIS. Quality has been unchanged: This version presented the qualification of the BAL HZG OLCI processing chain.

VI QUALITY CHANGES SINCE PREVIOUS VERSION

Improvements in the November 2024 release:

- The EAN reported in this version are not directly comparable to the previous document as new in-situ data was used.
- No changes or validation of the radiometric products were carried out in this version
- The performance of the updated 3 elements ensemble algorithm (Chl-a_{ENS3_CDF}) show a Determination coefficient of 0.2312 and RPD and APD of -2.8 and 66.1% for the MY multi sensor datasets and a Determination coefficient of 0.324 and RPD and APD of 4 and 51.2% for the NRT ad MY OLCI 300m datasets.
- On the same matchup data, for both the 1 Km MY multi sensor and the OLCI 300m datasets, there was a marked improvement on the previous version of the 3 elements ensemble algorithm (Chl-a_{ENS3}) (multi sensor: r^2 of 0.312 and RPD and APD of 13.8 and 73.3%; OLCI: r^2 of 0.324 and RPD and APD of 4 and 60.2%).

Improvements in the November 2022 release:

- The EAN reported in this version are not directly comparable to the previous version as new in-situ data was used.
- EUMETSAT v3.0.1 reprocessing for OLCI A and B
- The 3 elements ensemble algorithm (Chl-a_{ENS3}) was characterized by the lowest RMSD, RPD and APD (0.338, 40.9% and 89.9%), a marked improvement on the 6-bands MLP chlorophyll algorithm (Chl-a_{MLP_6b}) introduced in version 2.0 (RMSD, RPD and APD of 0.41,248.9% 117.9%)
- The results of the Pitarch et al. (2016) chlorophyll algorithm retired in version 2.0 show the highest RMSD *(0.52*) as well as RPD and APD (243 and 292%).

Improvements in the July 2022 release:

- The July-2022 release only concerns adoption of a new CMEMS nomenclature for products and datasets. Quality has been unchanged.

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